

BIO-DATA

1. Name and full correspondence address: **Dr. Pranay Abhimanue Bagde**
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3. Institution : Government College of Engineering Chandrapur. Maharashtra
4. Date of Birth :18/04/1988
5. Gender (M/F/T) : Male
6. Category Gen/SC/ST/OBC : SC
7. Whether differently abled (Yes/No) : No
8. Academic Qualification (Undergraduate Onwards)

S.No.	Degree	Year	Subject	University/Institution	% of marks
1.	B.Tech	2010	Metallurgy and Materials Science Engineering	College of Engineering Pune (COEP)	63.10%
2.	M.Tech	2012	Metallurgical and Materials Engineering	Indian Institute of Technology Kharagpur	68.70%
3.	PhD	2019	Metallurgical and Materials Engineering	Visvesvaraya National Institute of Technology , Nagpur (VNIT)	75 %

9. Ph.D thesis title, Guide's Name, Institute/Organization/University, Year of Award.

Tribological Behaviour of Plasma Sprayed $\text{Al}_2\text{O}_3\text{-TiO}_2$ and $\text{Cr}_2\text{O}_3\text{-TiO}_2$ Coatings under the guidance of Dr. S.G.Sapate and Dr. R.K.Khatirkar from Department of Metallurgical and Materials Engineering , Visvesvaraya National Institute of Technology , Nagpur (VNIT) in January 2019.

10. Recognition as Ph.D Guide/Supervisor in Mechanical Engineering under the Faculty of Science and Technology from 02/12/2021(**Recognition No: GUG/Ph.D.Sup.Rec./891/2021**)
11. Topics of Research/ Interest: Tribology of Engineering Materials, Microstructural Characterization, Development And Processing of HVOF And Plasma Sprayed Coatings
12. Courses taught:
Undergraduate : Materials Engineering (III SEM)
Manufacturing Processes(V SEM)

Post Graduate : Advanced Engineering Materials (I SEM)
Fracture Mechanics (II SEM)

13. Number of Ph.D completed/ongoing : 02 (Ongoing) details as follows:

S.No	Name of student	Title	Registration date	Register under University
1	Mr. Nitin Kumbhare	Experimental investigations for thermal performance of solar air heater by modeling absorber shapes	18/06/2022	Gondwana
2	Mr. Chakresh Shende	Investigation on Tribological response of plasma sprayed ceramic coating	RRC pending	Gondwana

14. Work experience (in chronological order).

S.No.	Positions held	Name of the Institute	From	To	Pay Scale
1	Assistant Professor	College of Engineering Pune (COEP)	12/08/2014	10/12/2015	15600-39100
2	Assistant Professor	Government College of Engineering Chandrapur	18/04/1988	Till date	Basic pay of 79900/-

15. Memberships of Professional / Non Professional Bodies: Tribology Society of India (Life Member)

16. Publications (*List of papers published in SCI Journals, in year wise descending order*).

S.No.	Author(s)	Title	Name of Journal	Volume	Page	Year
1	Sanket Mehar, SG Sapate, Nitesh Vashishtha, Avishkar Rathod, Pranay Bagde	Tribological performance of plasma sprayed Al ₂ O ₃ -TiO ₂ -ZrO ₂ ceramic coating	Materials Today: Proceedings (Elsevier)	45	4737-4741	2021
2	Sanket Mehar, SG Sapate, Nitesh Vashishtha, Pranay Bagde , Avishkar Rathod	Friction energy map for sliding wear of Al ₂ O ₃ -TiO ₂ and Cr ₂ O ₃ -TiO ₂ coatings	Surface Topography: Metrology and Properties(IOP Publishing)	9(1)	015020	2021
3	Sanket Mehar, SG Sapate, Nitesh Vashishtha, Pranay Bagde	Effect of Y ₂ O ₃ addition on tribological properties of plasma sprayed Al ₂ O ₃ -13% TiO ₂ coating	Ceramics International (Elsevier)	46(8)	11799-11810	2020
4	Sanket Mehar, SG Sapate, N Vashishtha, Pranay Bagde	Role Of Friction Energy And Tribo-Reduction In Abrasive Wear Of Alumina-Titania Coatings	Surface Review and Letters (SRL) (World Scientific Publishing Co. Pte. Ltd.)	27(12)	1-16	2020
5	Sanket Mehar, SG Sapate, Nitesh Vashishtha, A Rathod, Pranay Bagde	Investigation on structural, mechanical and tribological properties of plasma sprayed Al ₂ O ₃ -40% TiO ₂ coating	Materials Today: Proceedings (Elsevier)	38 (part 5)	2525-2531	2020
6	Ayush Kumar Goyal, SG Sapate, S Mehar, N Vashishtha, Pranay Bagde , A Rathod	Tribological properties of HVOF sprayed WC-Cr ₃ C ₂ -Ni coating	Materials Research Express (IOP Publishing)	6(10)	106415	2019
7	Nitesh Vashishtha, SG Sapate, JS Gahlot, Pranay Bagde	Effect of tribo-oxidation on friction and wear behaviour of HVOF sprayed WC-10Co-4Cr coating	Tribology Letters (Springer US)	66(2)	1-19	2018
8	Pranay Bagde , SG Sapate, RK Khatirkar, Nitesh Vashishtha	Friction and abrasive wear behaviour of Al ₂ O ₃ -13TiO ₂ and Al ₂ O ₃ -13TiO ₂ + Ni Graphite coatings	Tribology International (Elsevier)	121	353-372	2018

9	Pranay Bagde , SG Sapate, RK Khatirkar, Nitesh Vashishtha, Satish Tailor	Friction and wear behaviour of plasma sprayed Cr ₂ O ₃ -TiO ₂ coating	Materials Research Express (IOP Publishing)	5(2)	026410	2018
10	Nitesh Vashishtha, SG Sapate, Pranay Bagde , AB Rathod	Effect of heat treatment on friction and abrasive wear behaviour of WC-12Co and Cr ₃ C ₂ -25NiCr coatings	Tribology International (Elsevier)	118	381-399	2018
11	Nitesh Vashishtha, SG Sapate, Bikram Jyoti Sahariah, Pranay Bagde	Microstructural characterization and wear behaviour of high velocity oxy-fuel sprayed Cr ₃ C ₂ -25NiCr coating	Materials Today: Proceedings (Elsevier)	5(9)	17686-17693	2018
12	Pranay Bagde , Shubham Nandurkar, Tirthraj Lonare, Vaishnavi Fulzele	Design and fabrication of vertical axis wind turbine with magnetic repulsion	International Journal of Engineering Research and Technology	6(5)	790-795	2017
13	Pranay Bagde , Sanjay Sapate, Sanket Mehar, Avishkar Rathod	Effect of graphite addition on tribological behaviour of plasma sprayed Cr ₂ O ₃ -TiO ₂ coating	Materials Today: Proceedings (Elsevier)	56	2365-2370	2021

17. International Conferences Attended

- **Pranay bagde** , Rajesh Khatirkar, Sanjay Sapate. Study of structural characterization and wear properties of H11 steels. Meeting: materials science & technology 2013, Montreal, Canada(oral presentation). Funded by TEQIP
- **Pranay bagde** , Rajesh Khatirkar, Sanjay Sapate. Abrasive wear behavior of thermally sprayed (HVOF) die steel. 12th international surface engineering, paints & coating symposium and expo 2015 from 7th – 9th October 2015. (Poster presentation)

16.

Courses/Training attended	• NRC-M Workshop on Advances in Corrosion Engineering at Department of Materials Engineering, Indian Institute of Science, Bangalore from 22-24 Dec 2014. • UGC NRC-M workshop - Principles and Techniques of X-Ray Diffraction at Department of Materials Engineering, Indian Institute of Science, Bangalore from 10th to 21st June 2013 • QIP short term course on "Materials Tribology", 23-29 March, 2017, IIT-BHU, Varanasi. • GIAN course titled “Environmental Degradation of Materials in Power Industry and its Control” from Dec 11-17, 2017 at NIT Trichy, Tamilnadu. • Course on ‘Thermal Spray’ at ARCI Hyderabad India from 22-23 November 2014 under the guidance of Prof. Christopher Berndt. • Short term course on ‘characterization of materials’ from 22-24 March 2013 VNIT Nagpur. • Short term course on ‘Heat treatment and Metallography’, from 23-24 Feb. 2013 at VNIT, Nagpur. • Residential training program on " Worklife balance and stress management"(Program code - NPC/RD-D/HRM/TRG/07/2016- 17) from 17 to 21 October 2016 at Gangtok, Sikkim. • 6th Summer School in Tribology, 23 - 27 June 2014 at Indian Oil Institute of Petroleum Management (IIPM), Gurgaon. • TEQIP Sponsored Workshop on “Failure analysis of Engineering Materials” organized by Department of Metallurgical and Materials Engineering, VNIT, Nagpur in collaboration with Indian Society of Nondestructive Testing and Indian Institute of Metals (Nagpur Chapter) April 18th-20th, 2013. • One week Workshop on " Computational Fluid Dynamics and Software Training" from 13-17 July 2015 at VNIT Nagpur
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	• GIAN Course Titled "Environmental Degradation Of Materials In Power Industry And Its Control" Conducted by MME department, NIT Tiruchirapalli from Dec 11-17, 2017 • AICTE Sponsored one Week Faculty Development Program (FDP) on “Outcome Based Education: A Step Towards Excellence, Organizer: Government College of Engineering, Karad under "AICTE-MARGDARSHAN SCHEME" May 11-15, 2020 • DTE Sponsored One Week Short Term Training Programme on "Entrepreneurship, Employability & Life Skills" at Government college of Engineering Nagpur from 09th December-13th December 2019 • Two weeks Industrial training on " CAD and Additive Manufacturing: organized by CIIT , Government College of Engineering Chandrapur during September 20- 01 October 2021 • Participated in GIAN program on “Fatigue of Composite Materials” held at Dr B R Ambedkar National Institute of Technology, Jalandhar (Punjab) during 11-15 November 2019. • One Week Online FDP “Innovation, Entrepreneurship and its Relevance to Industry 4.0 practices in post Covid-19 Situations” organized by Terna Engineering College Nerul, Navi Mumbai from (25th - 29th May, 2020) • Online Training on "Change Management and Capacity Building for e-Governance" organized by Yashada Pune from 22nd to 24th December 2021 • Participated at 14th Summer School in Tribology held in the hybrid mode from 14-18 June 2022 at IndianOil Institute of Petroleum Management (IiPM) Gurugram
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